

Chapter 8

Detective Wireshark

I know what you did on the Internet

A promotional image for the TV show Mr. Robot. On the left, Rami Malek as Elliot Alderson is shown from the chest up, wearing a black zip-up hoodie. He is looking off to the side with a serious expression, his hands raised behind his head. The background is a solid, vibrant red. To the right of the image, the text "RED TEAM." is written in white, bold, sans-serif capital letters.

RED TEAM.

A promotional image for the TV show Mr. Robot. On the right, David Finck as Mr. Robot is shown from the chest up. He is wearing a dark suit jacket over a light blue button-down shirt. He has his arms crossed and is looking directly at the camera with a slight, enigmatic smile. The background is a bright blue sky with some light clouds. Above the image, the text "BLUE TEAM" is written in white, bold, sans-serif capital letters.

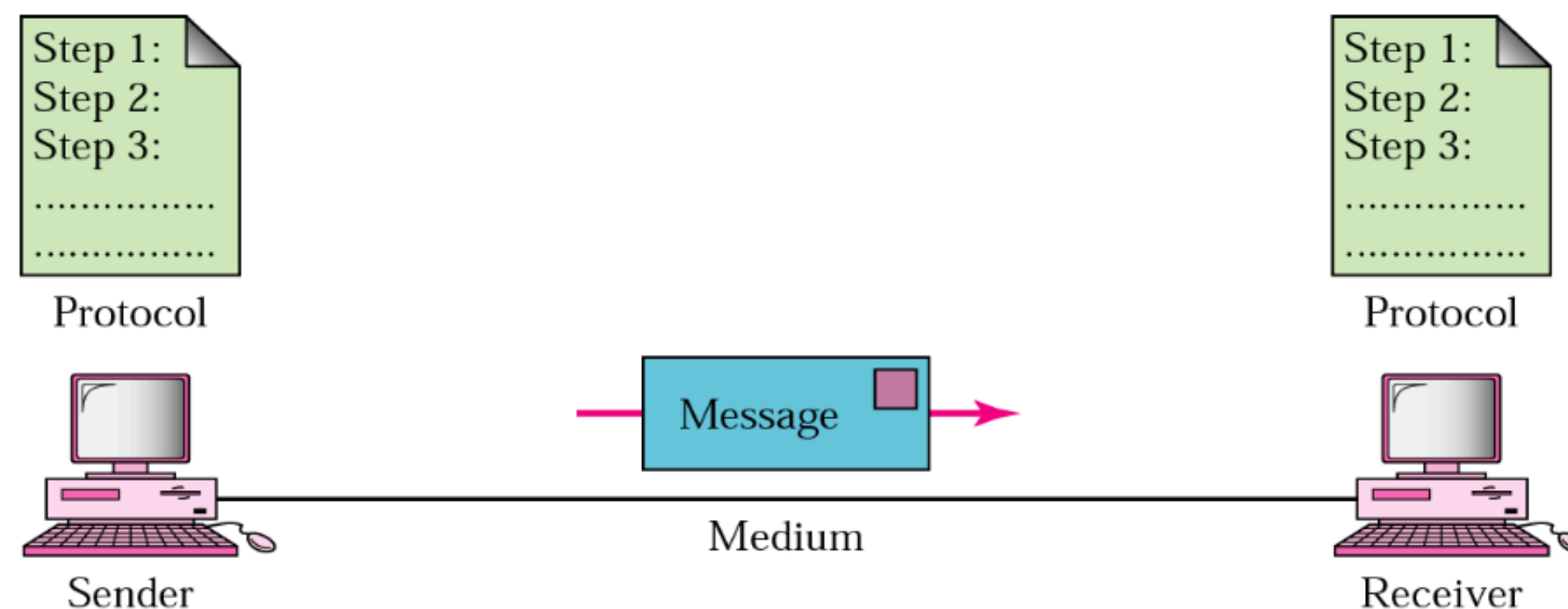
BLUE TEAM

MR. ROBOT

HOW IS DATA TRANSMITTED?

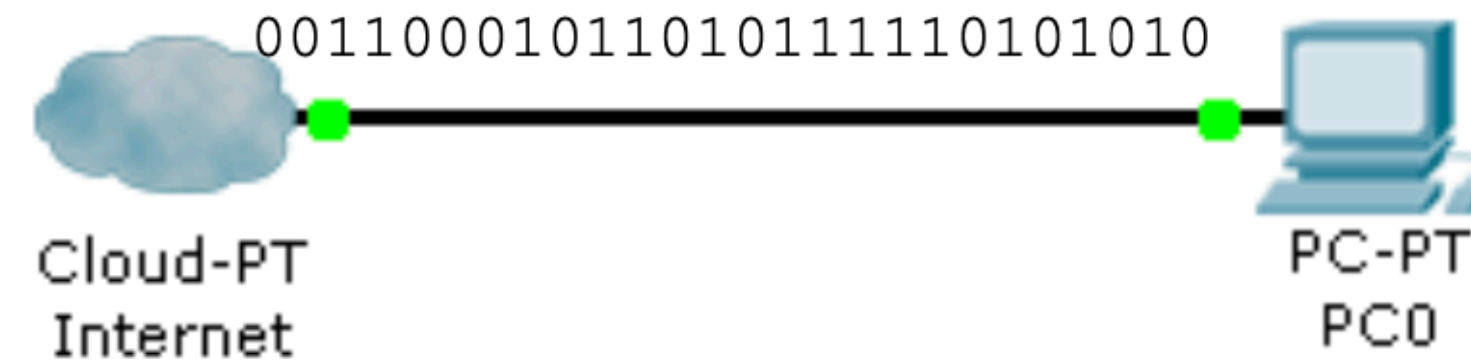
5 components of Data Comm

- **Message**
 - Information (data) to be communicated, (e.g. text, number, pictures, sound, video, etc)
- **Sender**
 - Device that sends the message, (e.g. computer, telephone, etc)
 - Message generator (computer),
 - and intermediate sender (relay the message, e.g. router)
- **Receiver**
 - Device that receive the message, (e.g computer, telephone, etc)
 - End system (computer), and intermediate receiver (e.g. router, repeater)
- **Medium**
 - Physical path by which a message travels from sender to receiver, (e.g. wire, cable, fiber, radio wave)
- **Protocol (the hardest to understand)** since it is rather “abstract”.

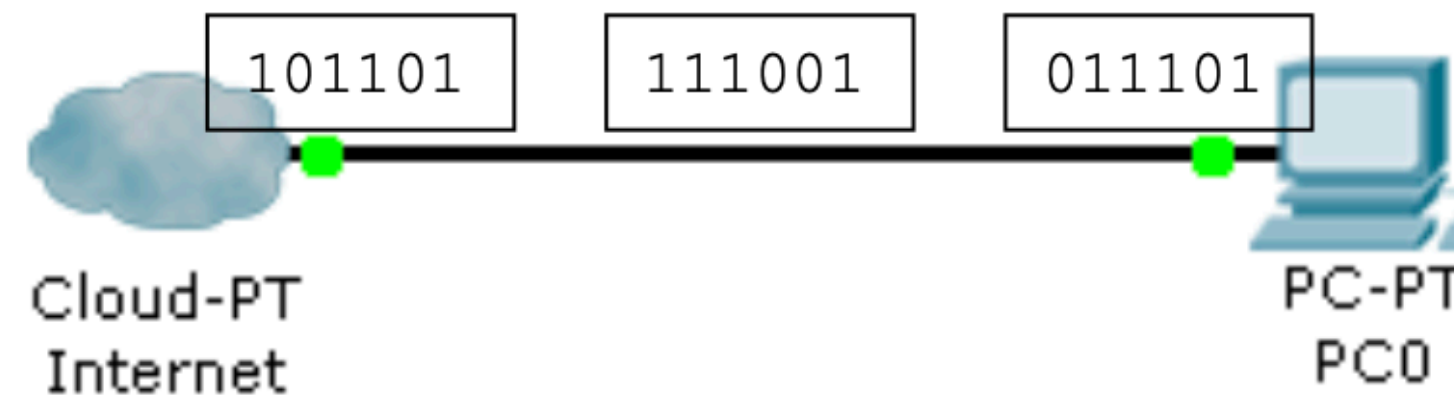


Data Transmission in Packets

- In computer networks, data is NOT sent a in continuous stream.

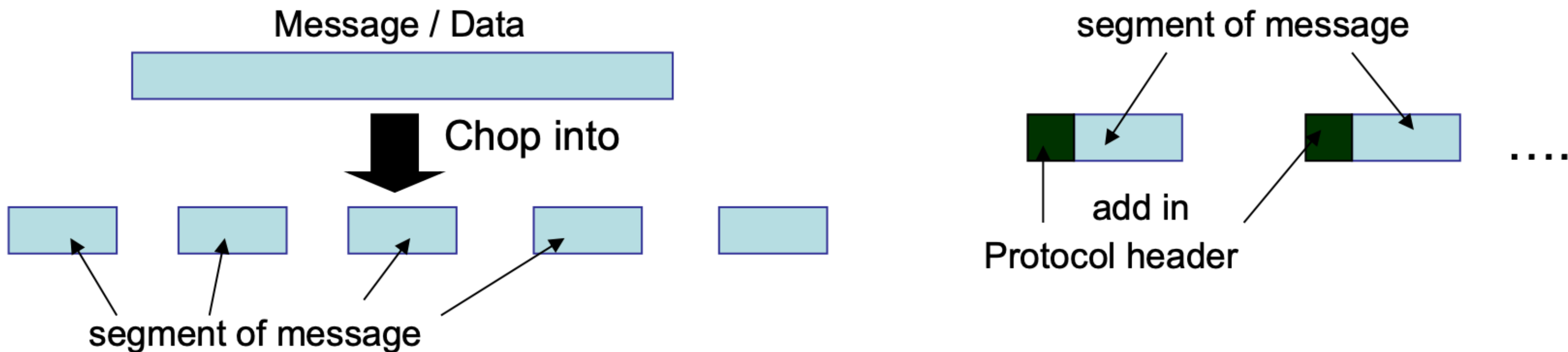


- In computer networks, data is sent a in packetized form.
 - Meaning chunk by chunk (or packet by packet)

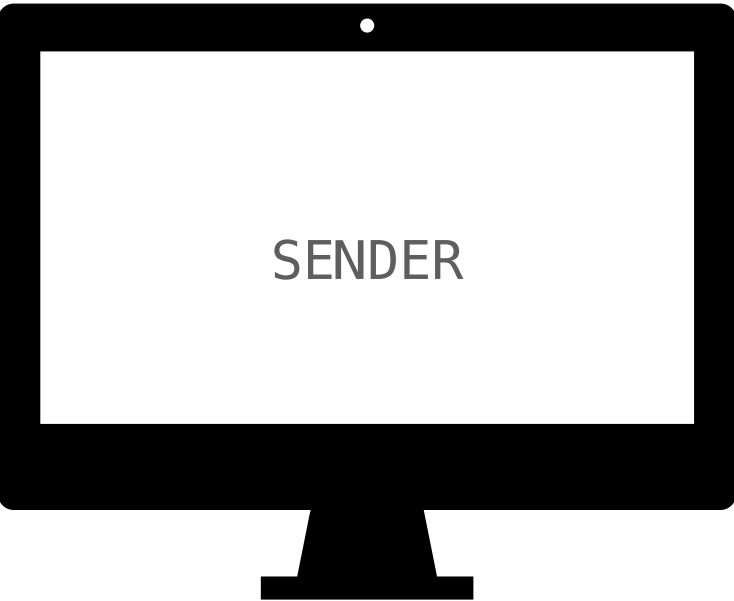
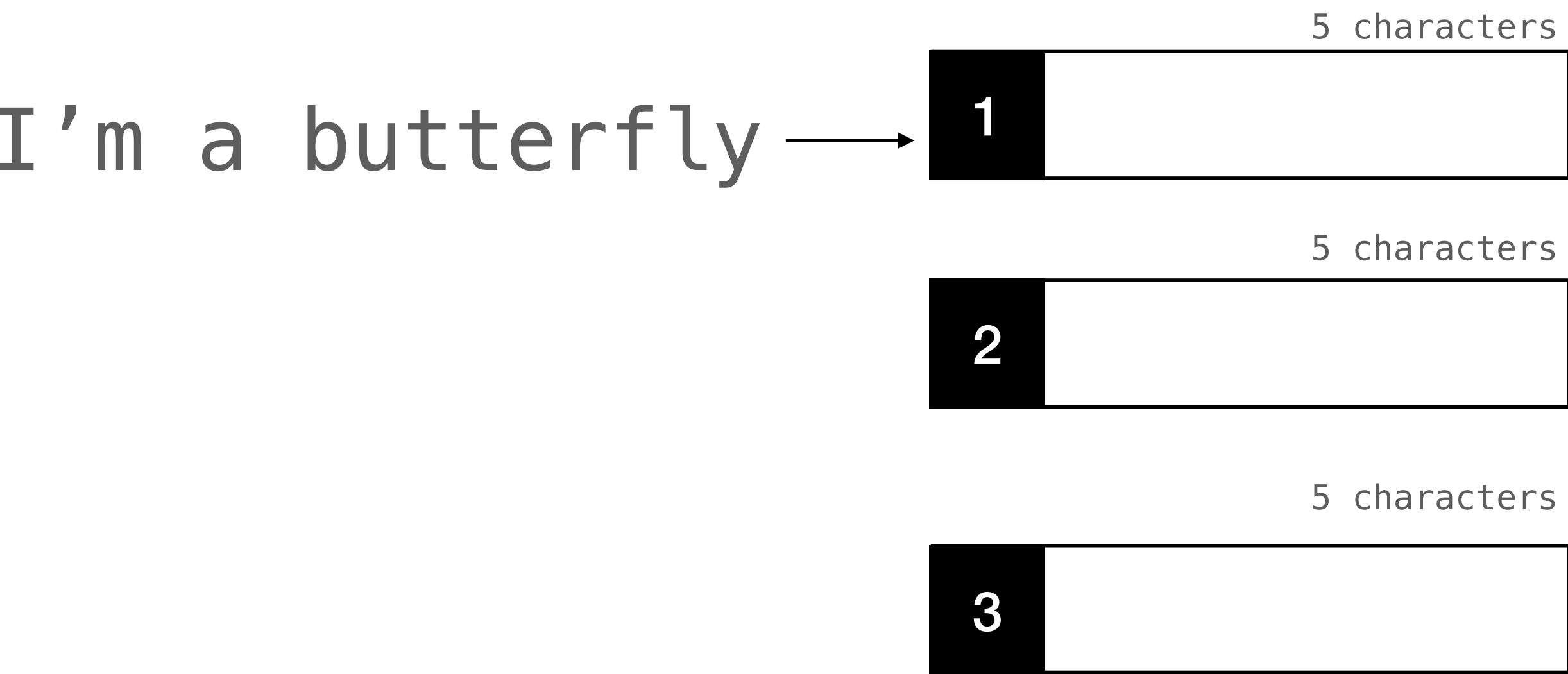


Segmentation of Data & Protocol Header

- Data (or a file) is first being segmented (or chopped into pieces) before sent out.
- After the segmentation (or chopping), protocol headers have been added at the beginning of each data segment.
- In the “header” it normally contain
 - Management information for the network
 - More importantly, the addresses (e.g: IP address, port number, etc)

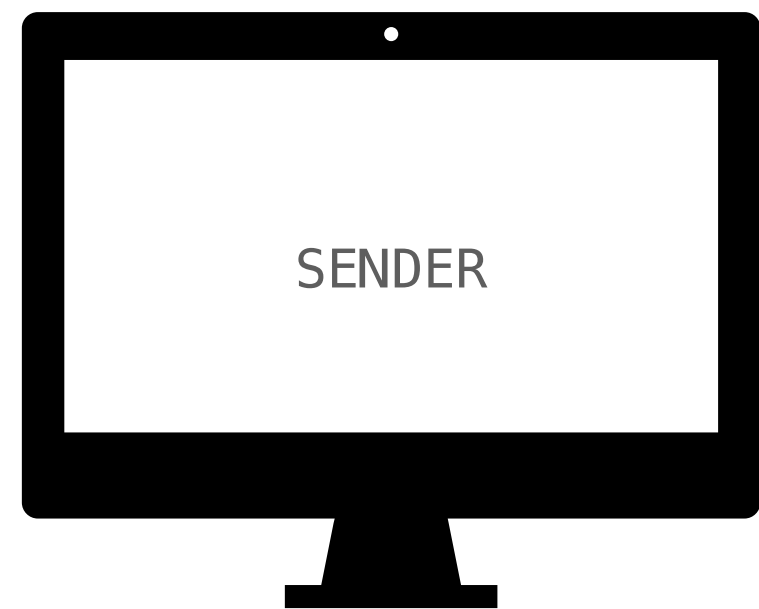


Sending and receiving data

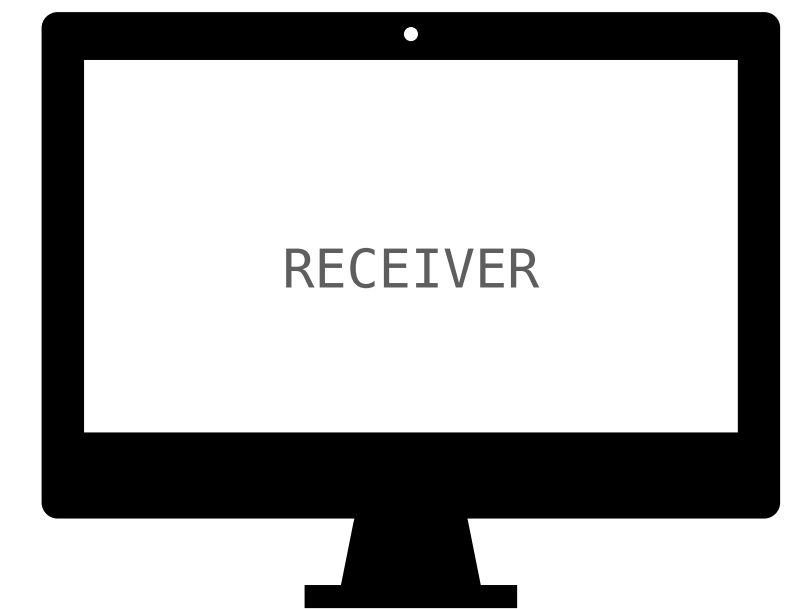
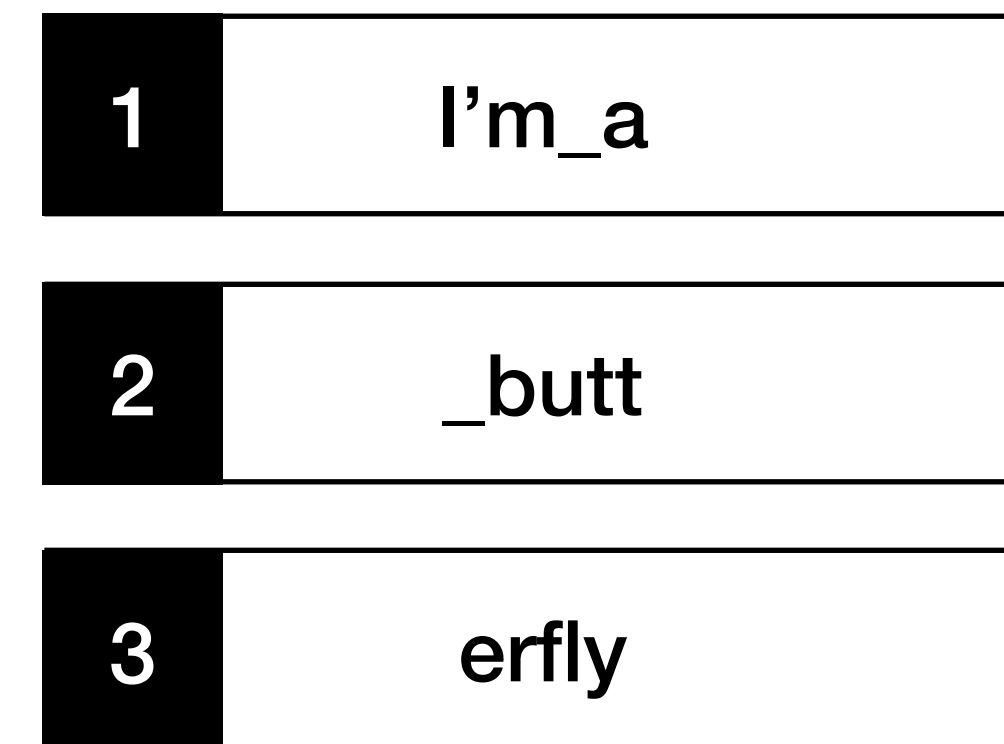


Sending and receiving data (with TCP)

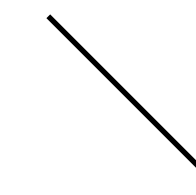
TCP is used to reorder the segment that may arrive out of order
TCP is also used to resend the segment if missing in delivery



SENDER



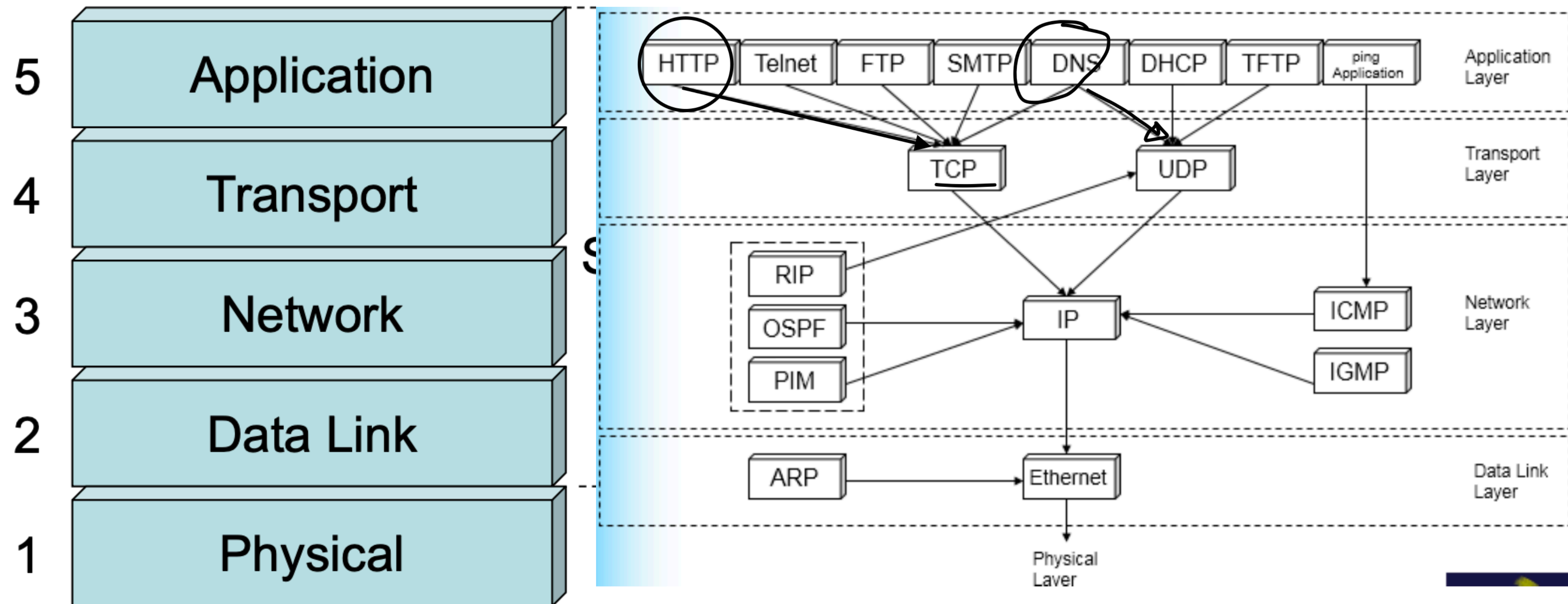
RECEIVER

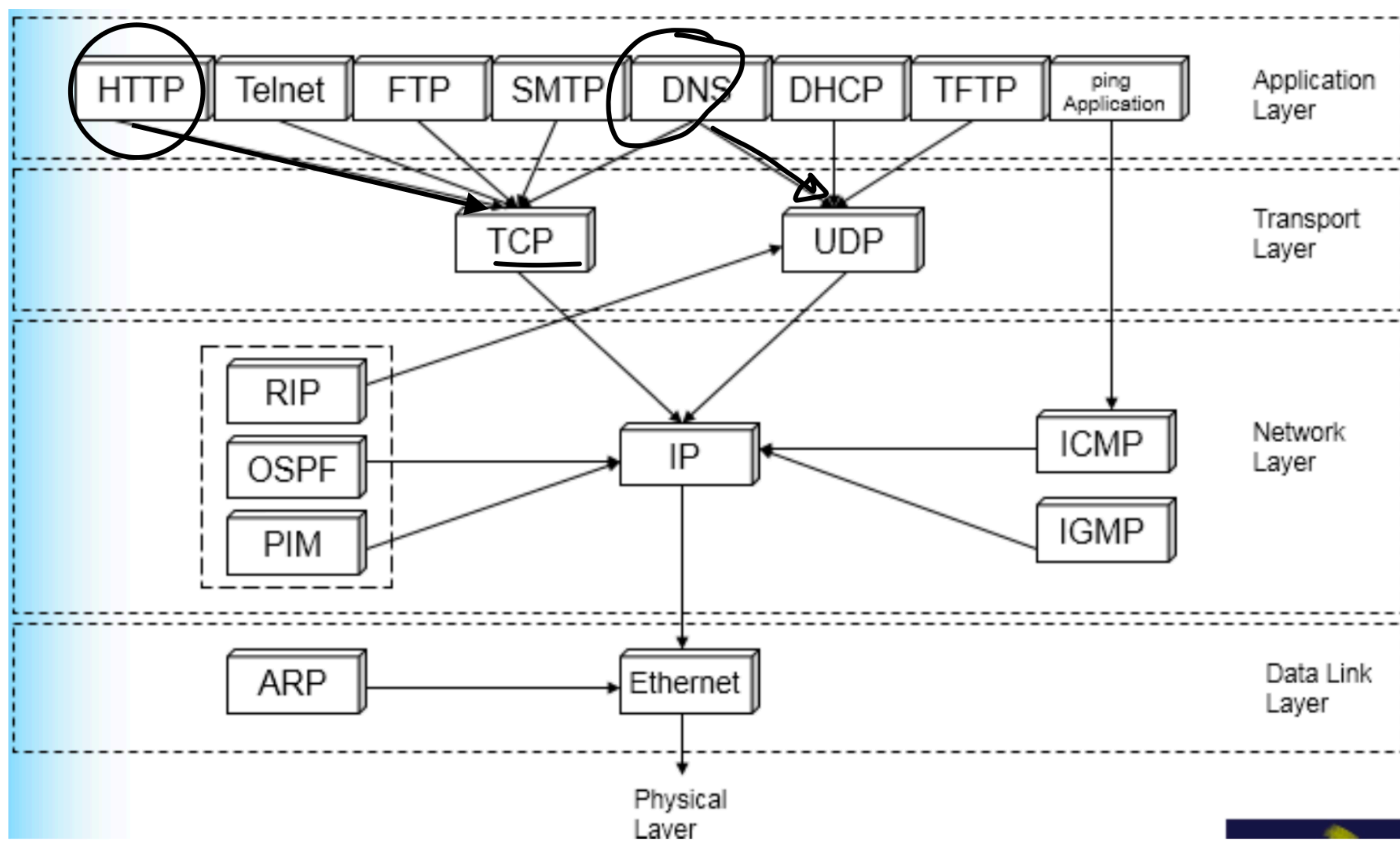


I'm a butterfly

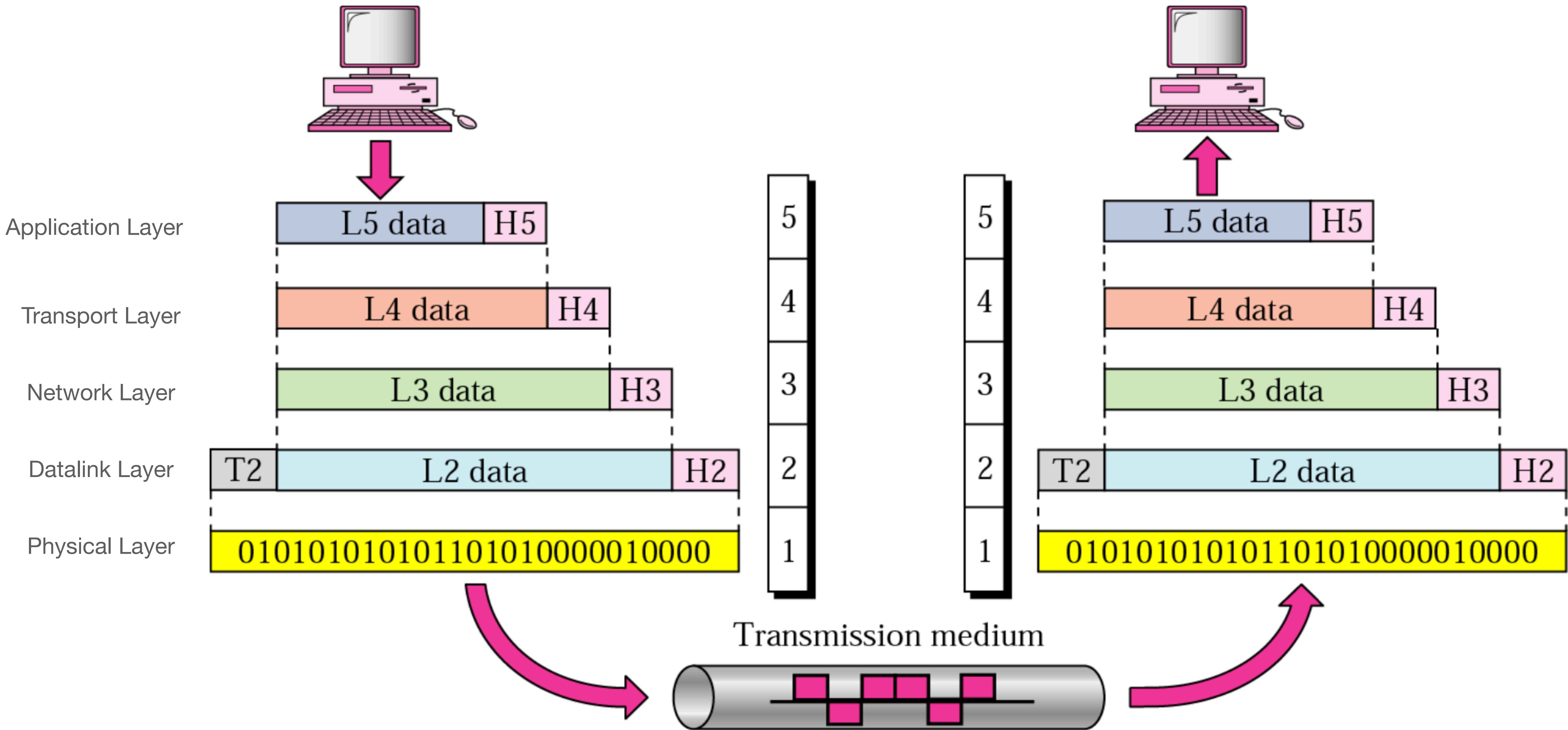
DATA ENCAPSULATION

The Internet Model

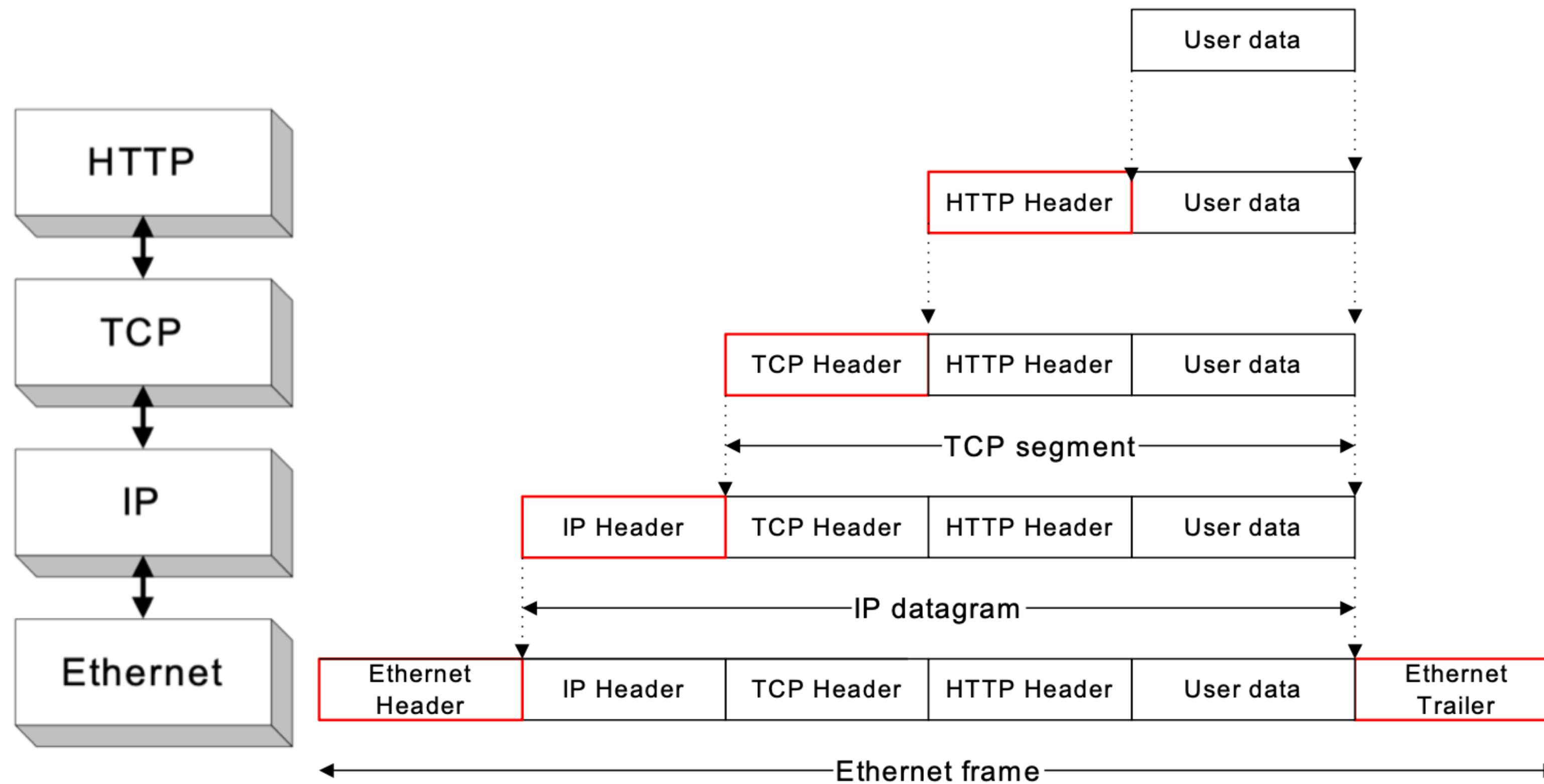




Data need to be packed into boxes before delivery to receiver
Packing here means adding a header for each different layers



- As data is moving down the protocol stack, each protocol is adding layer-specific control information



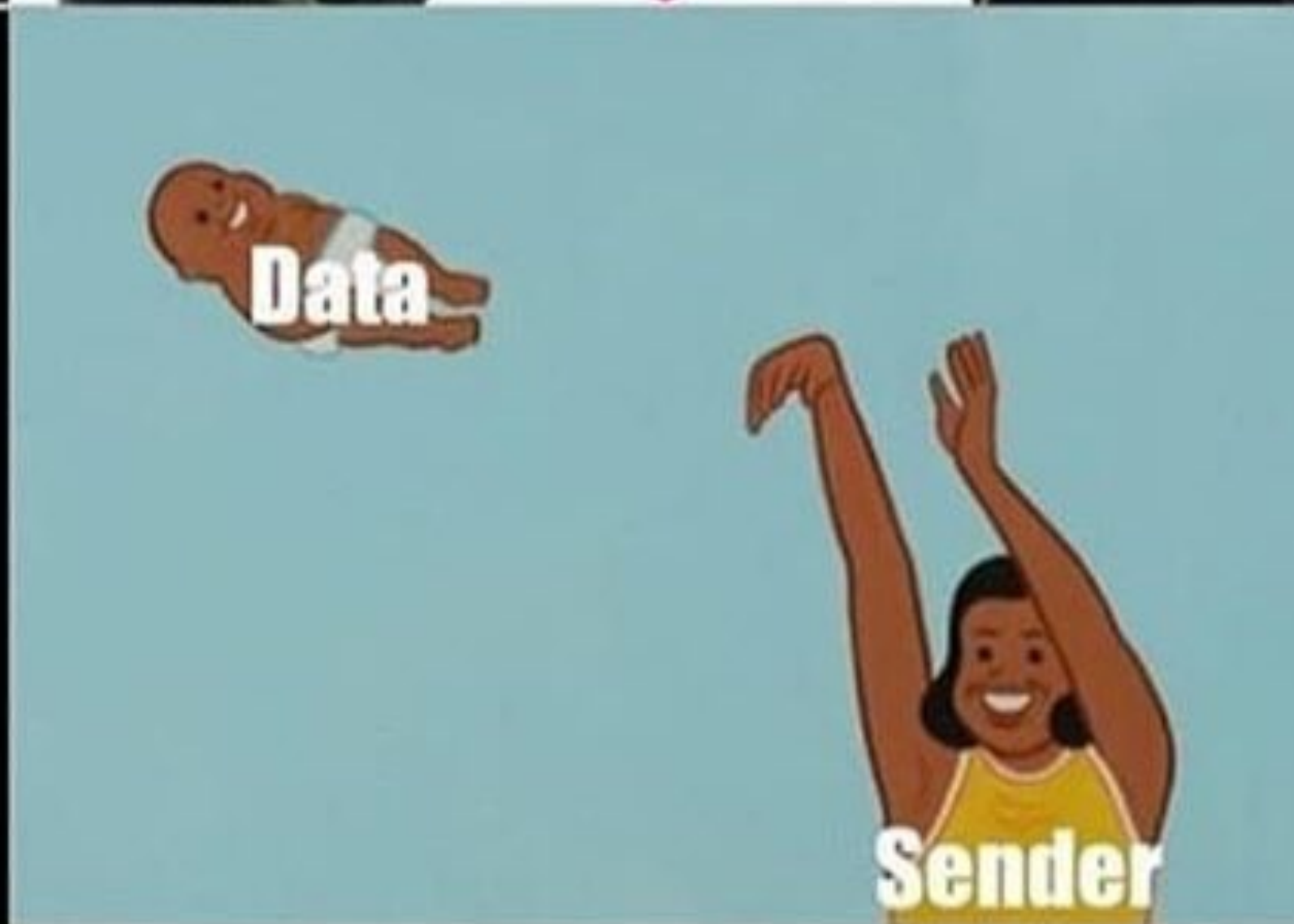
TCP vs UDP

Feature	TCP	UDP
Connection status	Requires an established connection to transmit data (connection should be closed once transmission is complete)	Connectionless protocol with no requirements for opening, maintaining, or terminating a connection
Data sequencing	Able to sequence	Unable to sequence
Guaranteed delivery	Can guarantee delivery of data to the destination router	Cannot guarantee delivery of data to the destination
Retransmission of data	Retransmission of lost packets is possible	No retransmission of lost packets
Error checking	Extensive error checking and acknowledgment of data	Basic error checking mechanism using checksums
Method of transfer	Data is read as a byte stream; messages are transmitted to segment boundaries	UDP packets with defined boundaries; sent individually and checked for integrity on arrival
Speed	Slower than UDP	Faster than TCP
Broadcasting	Does not support Broadcasting	Does support Broadcasting
Optimal use	Used by HTTPS, HTTP, SMTP, POP, FTP, etc	Video conferencing, streaming, DNS, VoIP, etc

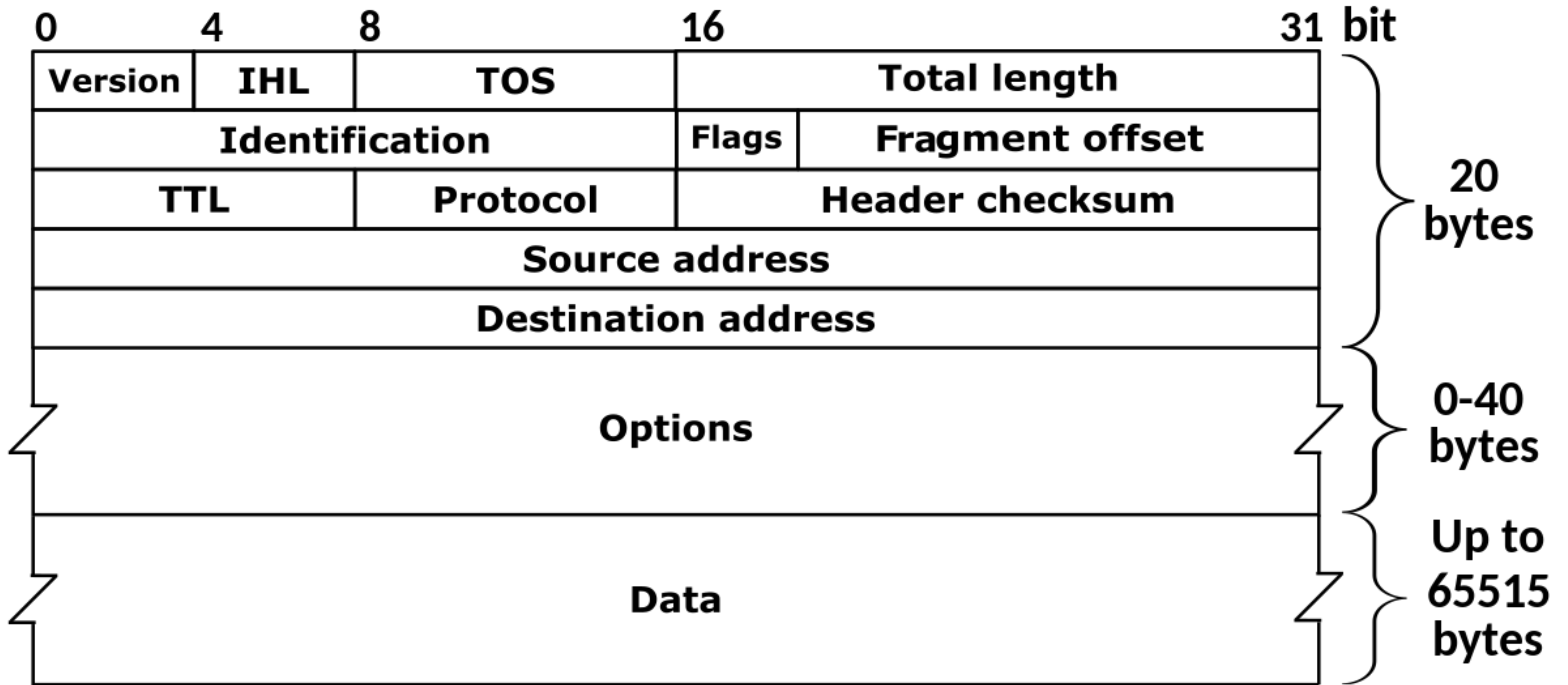
TCP



UDP



IPv4 Header



VERSION: Version of the IP protocol (4 bits), which is 4 for IPv4

HLEN: IP header length (4 bits), which is the number of 32 bit words in the header. The minimum value for this field is 5 and the maximum is 15.

Type of service: Low Delay, High Throughput, Reliability (8 bits)

Total Length: Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes.

Identification: Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits)

Flags: 3 flags of 1 bit each : reserved bit (must be zero), do not fragment flag, more fragments flag (same order)

Fragment Offset: Represents the number of Data Bytes ahead of the particular fragment in the particular Datagram. Specified in terms of number of 8 bytes, which has the maximum value of 65,528 bytes.

Time to live: Datagram's lifetime (8 bits), It prevents the datagram to loop through the network by restricting the number of Hops taken by a Packet before delivering to the Destination.

Protocol: Name of the protocol to which the data is to be passed (8 bits)

Header Checksum: 16 bits header checksum for checking errors in the datagram header

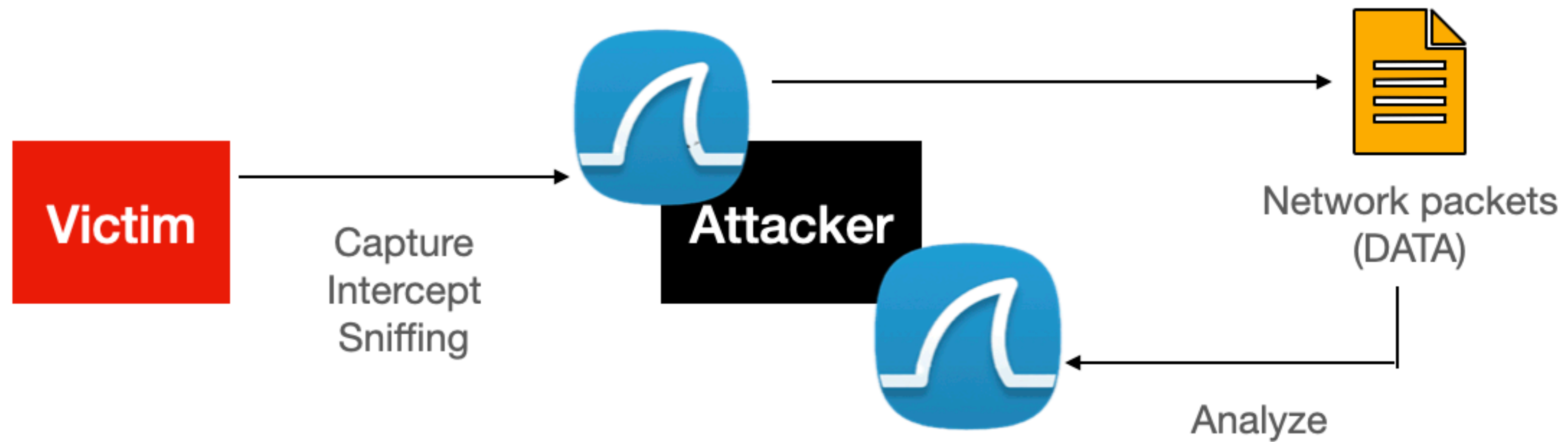
Source IP address: 32 bits IP address of the sender

Destination IP address: 32 bits IP address of the receiver

Option: Optional information such as source route, record route. Used by the Network administrator to check whether a path is working or not.

The flow of hacking

The (oversimplified) flow of Hacking



HOW TO CAPTURE DATA? (featuring PLASHSPEED router)

LATEST PLASH SPEED 5 WIFI 6 "ROUTER"!



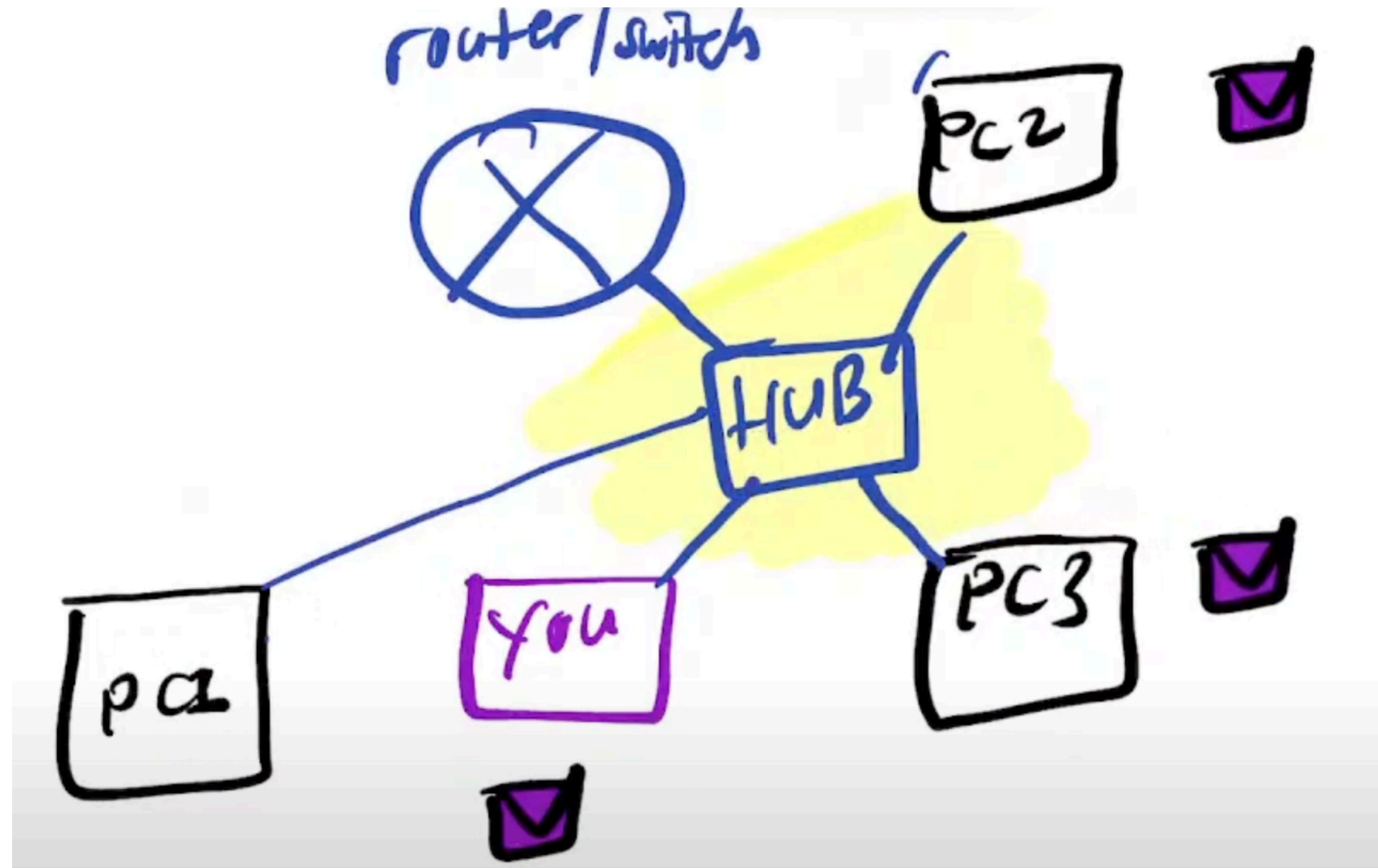
Free

"WiFi Adjuster"

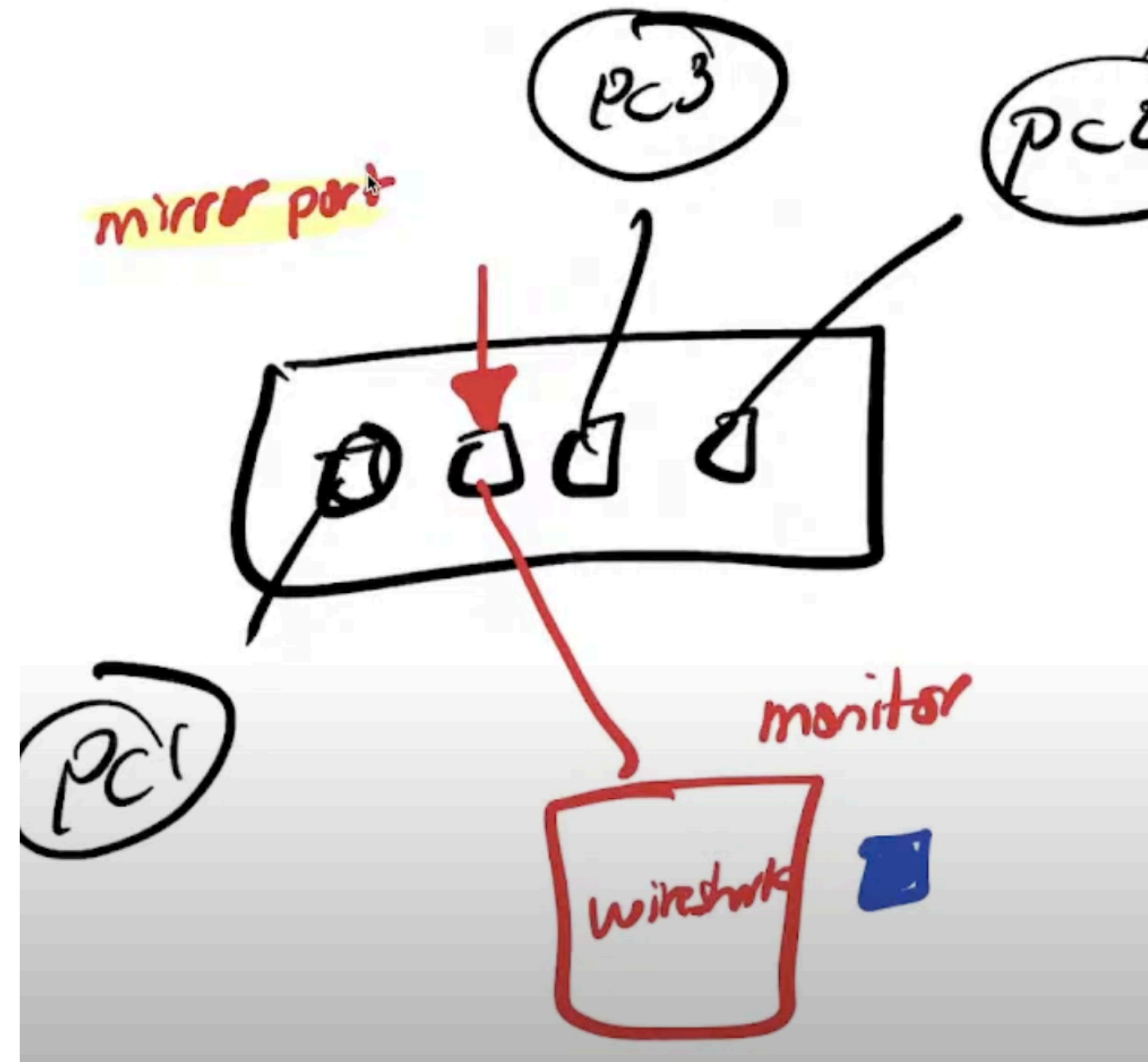
Not sure which to buy?



Method 1: Go cheap with Hub

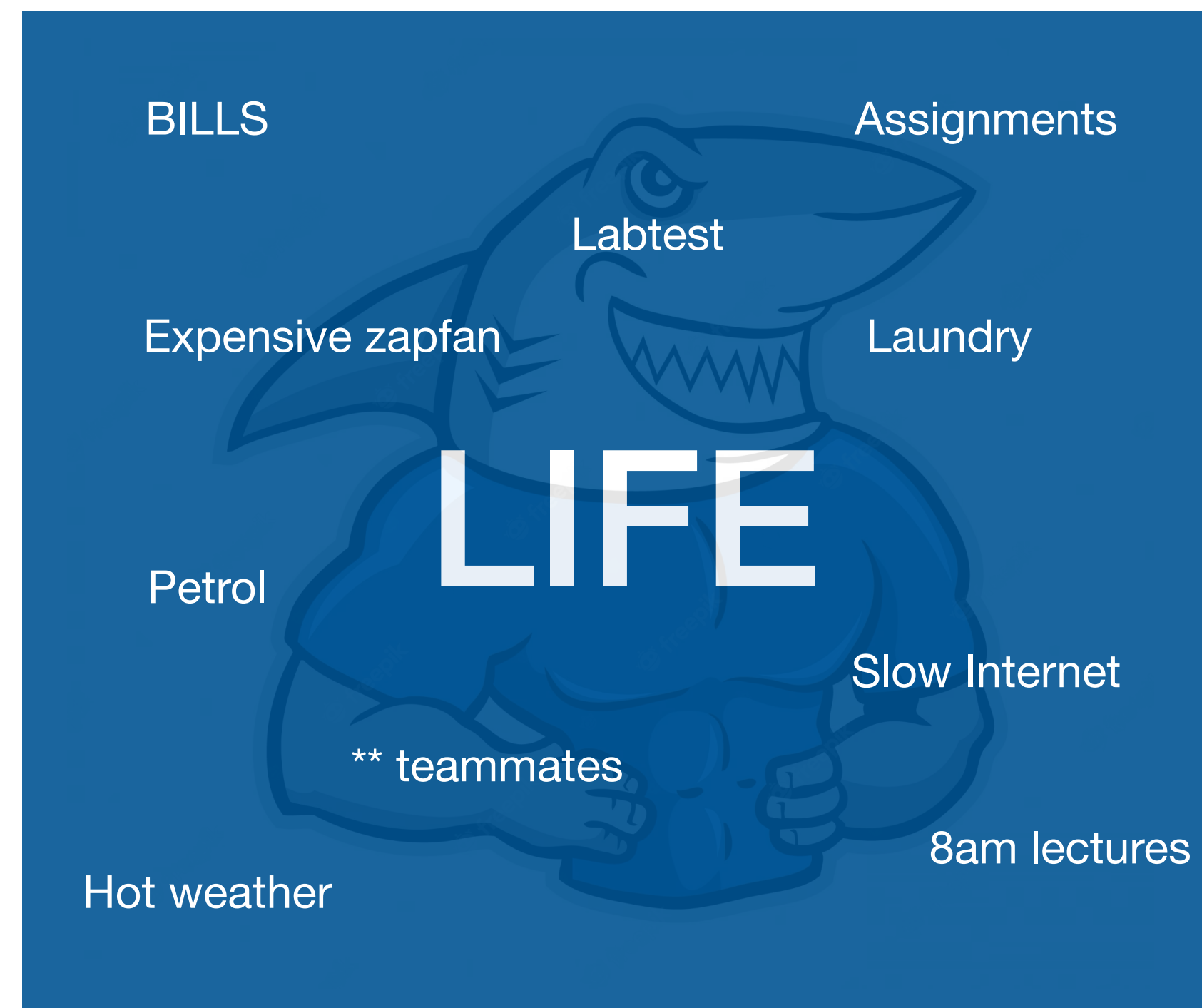


Method 2: Switch port mirroring



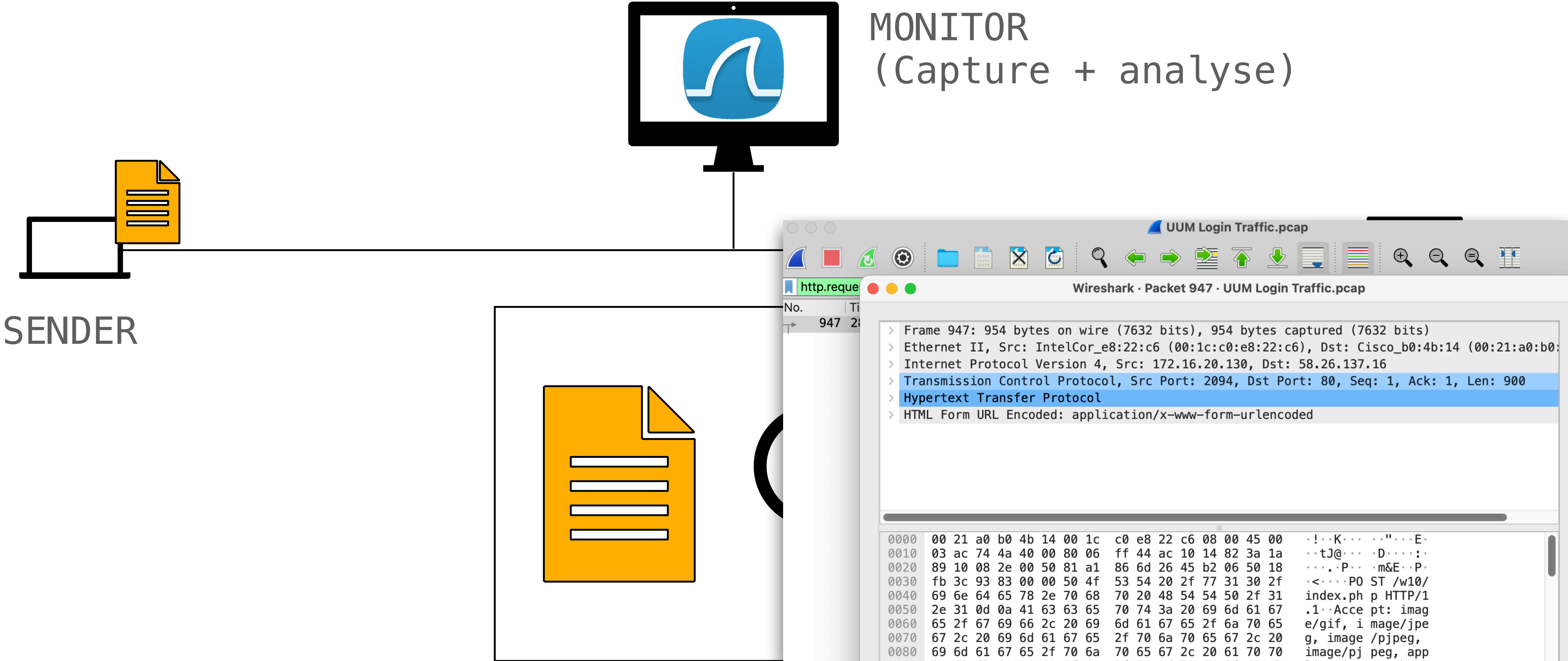
Wireshark

WIRESHARK



PEEKING INSIDE YOUR DATA?

What is inside your packets?



DETECTIVE WIRESHARK

find
server location

recover
user & pass

rebuild
missing data



Ocean Ave

```
* NTLM : 8846f7eae8fb117ad06bdd
tspkg :
* Username : Administrator
* Domain : WIN2008R2
* Password : password
wdigest :
* Username : Administrator
* Domain : WIN2008R2
* Password : password
kerberos :
* Username : Administrator
* Domain : WIN2008R2
* Password : password
```

10101010101010101
00000101010101010



Geo-IP (Wireshark)

GeoIP in Wireshark allows you to identify the geographic location of network hosts based on their IP addresses.

Since each data is geo-tagged. We can use Maxmind database to locate the actual server location. This is useful to detect the whereabouts of targets or attackers.

Full tutorial: <https://wiki.wireshark.org/HowToUseGeoIP.md>

HTTP Methods

We can use Wireshark to recover password transmitted in plaintext.

Look at http.post packet

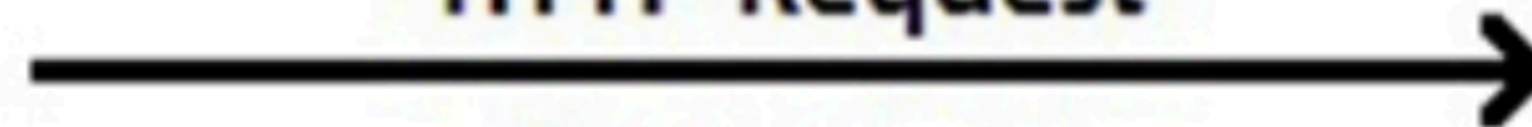
HTTP Client



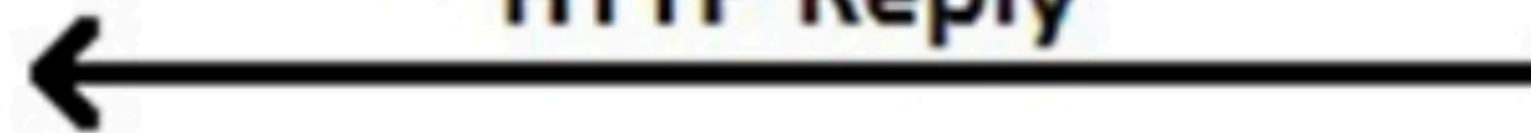
HTTP Server



HTTP Request



HTTP Reply



http POST
http GET
http PUT

YouTube LIVE
WEDNESDAY, 7pm-8pm
(Packet Analysis)
Very important